

SummarAI

Individual Reflection

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NLP Group Project · 2025–2026

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My Contributions

Frontend

I built the entire frontend as a single HTML file (`src/static/index.html`) with no framework dependencies — plain JavaScript and CSS only. The interface includes a drag-and-drop upload area, a results panel with the summary, color-coded takeaway bubbles (purple for decisions, grey for notes), an action item checklist, and a collapsible sidebar showing the timeline of saved summarizations per project. The layout is fully responsive and works on iPhone screens.

Loading state: Transcription can take several minutes, so the interface shows a spinner with a running elapsed timer. Results render progressively as the JSON comes back from the API.

Error handling: All failure states are handled explicitly — unsupported file type, missing API key, model error — so the user gets a clear message rather than a silent failure.

Failure Mode Analysis

After the evaluation was run, I documented the most informative failure cases in `data/eval/failure_analysis.md`. Each case includes the transcript excerpt, the model's output, the gold label, and a hypothesis for why it failed. Main patterns:

Whisper mishearing proper nouns: Causes the owner field to be wrong even when the task is correct.

Phantom action items: The model generates commitments from hypothetical discussions.

Zero output on non-meeting audio: Correct behavior for podcasts, but may confuse users.

Implied responsibility: When no person is named, the model leaves owner null rather than guessing.

Labeled Transcripts

I contributed approximately 8 labeled transcripts to `eval_set.jsonl`, deliberately designing difficult cases: meetings where no one is explicitly assigned a task, shared responsibilities, and informally stated deadlines.

What I Learned

The most important thing I took away is how NLP errors propagate through a pipeline. Whisper's transcription errors do not stay in the transcription layer — they flow directly into the LLM output. If Whisper mishears a name, the action item will have the wrong owner even if the model does everything else correctly. This cascading failure is hard to see from an aggregate F1 score alone, and it made me understand why failure mode analysis is treated as a required component of the assignment. I also learned what information extraction means as an NLP task — it sits between pure classification and open-ended generation, and the challenge is that the output structure is constrained but the surface form is open, which is exactly why simple rule-based approaches fail on real, messy transcripts.